# How Much Does Anticipation Matter for MPCs?

Evidence from Seasonal Bonuses and Lottery Wins

David Cesar Heymann (CaixaBank) and Charlie Nourse (LSE)

2026-10-19

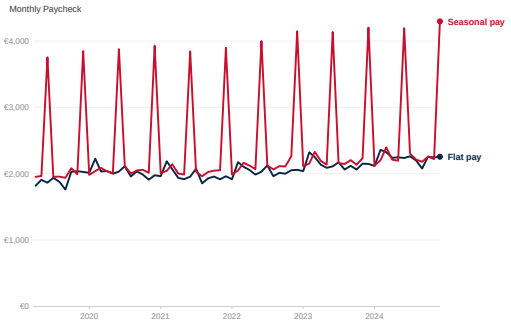
# Two Spanish Institutions

## El Gordo: Winners celebrate share in world's richest draw

22 December 2019

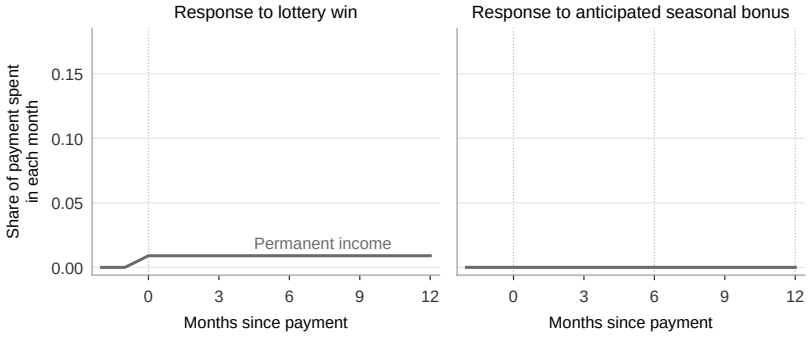



Thousands of people have been celebrating in Spain after winning a share of the €2.38bn ($2.64bn; £2.03bn) prize pot in the world's richest lottery.



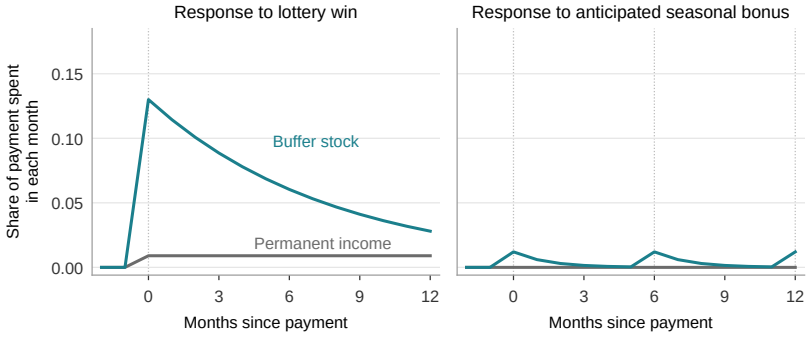

▸ Lotteries in our data ▸ Seasonal pay around the world

# Consumption theory makes strong predictions about these




**Permanent income hypothesis** (Friedman 1957; Hall 1978): forward-looking, unconstrained households smooth income changes.

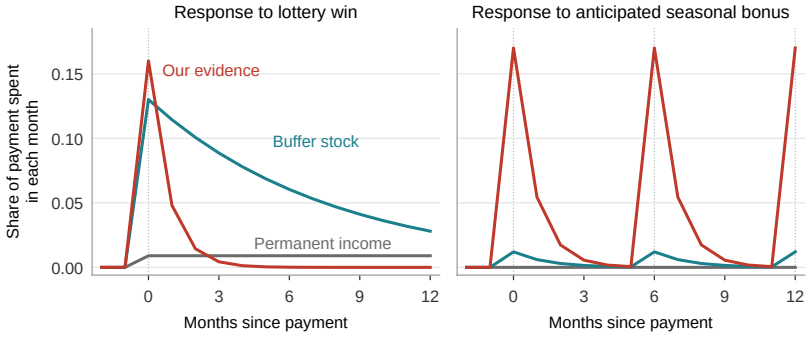
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## This paper

- Estimate MPCs from **fully anticipated, recurring, large** income changes.
- Decompose by category: broad-based, not driven by durables or payment timing.
- Also estimate MPCs from income **shocks** of comparable size.
- Demonstrate that standard models cannot match our findings.
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| | MPC from lottery win | | | MPC from anticipated bonus | | |
|---|---|---|---|---|---|---|
| | Size | For highest liquidity households | Persistence | Size | For highest liquidity households | Persistence |
| **Our findings** | 0.30 | >0.1 | Short and sharp | 0.21 | >0.1 | Short and sharp |

For the same households, the bonus and lottery MPCs are equal (both $\approx 0.25$).

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| Buffer-stock model | ✓ | × | × | × | × | × |
| Mental accounting model | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

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# Contribution: isolating the role of anticipation in MPCs

| Literature | Key papers | What we add |
|---|---|---|
| **MPCs and seasonal pay** | Souleles (1999), Johnson et al. (2006), Parker et al. (2013), Parker (2017), Kueng (2018), Baugh et al. (2021), Fagereng et al. (2021), Boehm et al. (2025); Paxson (1993), Browning & Collado (2001), Stephens & Unayama (2011), Hori & Shimizutani (2012) | Study income changes that are<br>• **fully anticipated, recurring, and large**<br>• with a **clean control group**<br>• and **adequate power**<br><br>Direct comparison with lottery MPCs **for the same households** |
| **Consumption theory** | Hall (1978), Deaton (1991), Carroll (1997), Kaplan & Violante (2014), Kaplan, Moll & Violante (2018), Auclert et al. (2025) | Seasonal bonuses in the buffer-stock model: a **sharper moment** for distinguishing behavioural and buffer-stock models |
| **Mental accounting** | Thaler (1985), Shefrin & Thaler (1988), Graham & McDowall (2025), Gimeno-Ribes (2025) | How mental accounting interacts with **anticipated bonuses**<br><br>Implications of mental accounting for **fiscal policy and debt relief** policies |

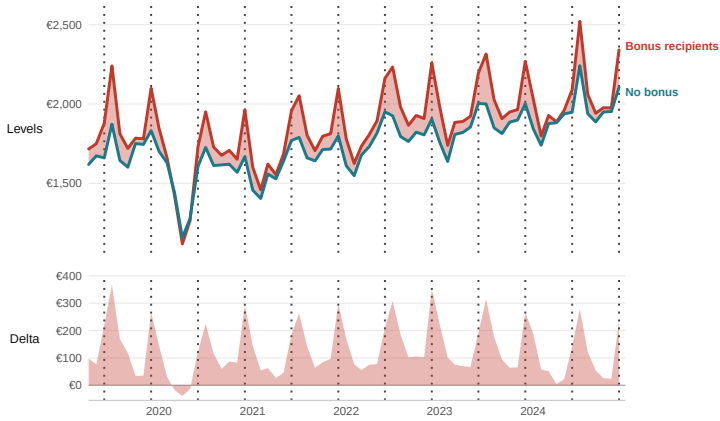
# Data: bank records for 250,000 Spanish households

- CaixaBank: the largest domestic bank, with 18.9m clients and 36% of Spanish paychecks.
- Monthly data, 2019–25.
  - For seasonal bonuses, panel of 250,000 households; about half on seasonal pay, half prorated.
  - For lotteries, panel of 30,000 lottery winners.
- Observe spending (with category breakdown), salary, current account balance, lottery wins.
- Total spending is the sum of:
  - card spending (65%)
  - essential bills (10%)
  - other bills (5%)
  - cash withdrawals (20%).
- Excludes bank transfers: rent, cars, large renovations.
- Validated against official surveys (EPF, ECV, EFF).

‣ Official data ‣ Main earners ‣ Spending ‣ Spending, 1 adult ‣ Salary ‣ Deposits

# Seasonal Bonus MPCs

# Spending diverges each and every bonus month




‣ Salaries ‣ Card spending ‣ Cash

# Identification: bonus status is set by collective bargaining agreements

- Six-month event windows around each bonus episode.
- Difference-in-differences with two-way fixed effects:

$$C_{i,t} = \alpha_i + \lambda_t + \sum_{s \neq -1} \tau_s D_{i,t-s} + u_{i,t}$$

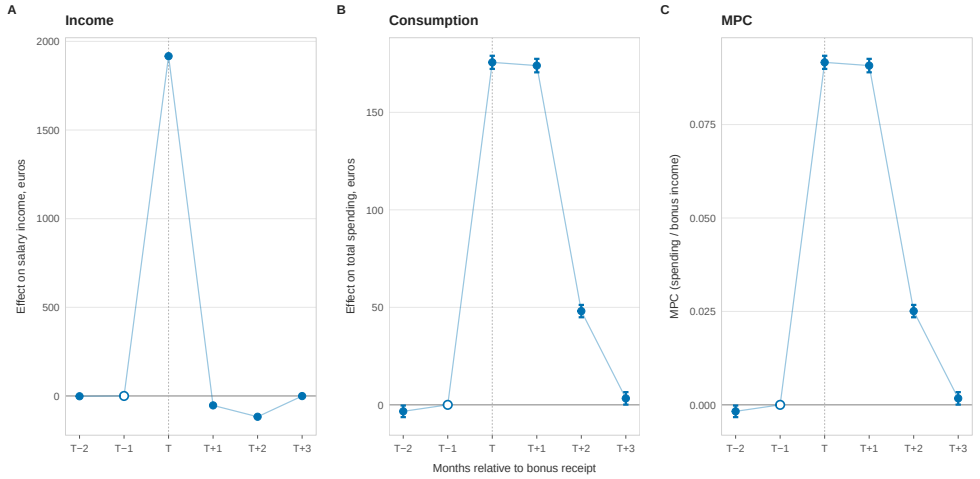
$C_{i,t}$: spending of household $i$ in month $t$; $D_{i,t-s} = 1$ if a bonus arrived $s$ months ago; base month $s = -1$.

Parallel trends?

- Bonus status set at firm level by collective bargaining agreements: workers don't choose.
- Visual comparison: ‣ Time series
- Group balance: ‣ Balance check
- Robustness: ‣ Conditional on covariates

‣ Anticipation effects ‣ All event studies

# 3-month MPC = 0.21, 1-month MPC = 0.09




▸ MPC calculation ▸ Event studies ▸ Income response ▸ Regression tables ▸ B&C (2001)

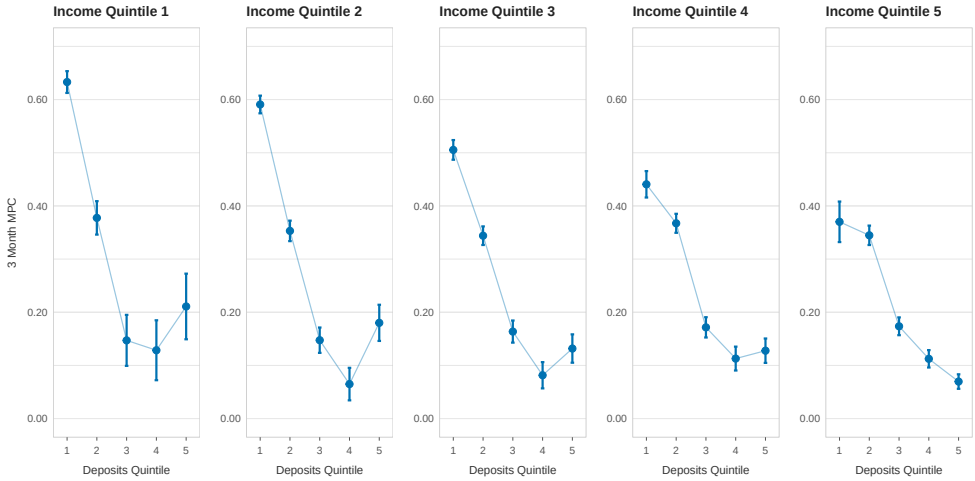
# 3-month spending response by category: the effect is broad-based

| Category | Effect | S.E. | Share (%) | Cumulative (%) |
|---|---|---|---|---|
| Cash withdrawals | 65.5*** | (5.6) | 20.7 | 20.7 |
| Food, drinks, tobacco | 40.6*** | (1.9) | 12.9 | 33.6 |
| Restaurants | 24.1*** | (1.6) | 7.6 | 41.2 |
| Other retail | 22.9*** | (2.3) | 7.2 | 48.4 |
| Fashion | 20.5*** | (2.0) | 6.5 | 54.9 |
| Insurance bills | 17.2*** | (5.7) | 5.4 | 60.3 |
| Other bills | 12.0* | (6.3) | 3.8 | 64.1 |
| Transport | 11.8*** | (2.5) | 3.7 | 67.9 |
| Utility bills | 10.0*** | (1.2) | 3.2 | 71.1 |
| Culture and shows | 7.3*** | (2.0) | 2.3 | 73.4 |
| Gas stations | 6.5*** | (0.7) | 2.1 | 75.4 |
| Furniture | 5.2*** | (1.3) | 1.6 | 77.1 |
| Hotels | 5.1*** | (1.7) | 1.6 | 78.7 |
| Travel agencies | 4.9*** | (1.8) | 1.6 | 80.2 |
| Home appliances, IT | 3.2** | (1.6) | 1.0 | 81.2 |
| Pharmacies | 2.3*** | (0.5) | 0.7 | 81.9 |
| Other essential goods | -0.6 | (1.7) | -0.2 | 81.7 |
| Residual | 57.7 | | 18.3 | |
| TOTAL (Total spending) | 316.2*** | (15.1) | 100.0 | |

Sum of coefficients at T, T+1, T+2. SEs clustered by household × event. * p<0.10, ** p<0.05, *** p<0.01.

▸ Larger categories ▸ Spending shares ▸ Lottery version ▸ Official data

# MPC heterogeneity over the joint distribution of income and liquidity




One covariate | Over time | Age | Covariates | Distributions | Quintiles | Cash

# Lottery MPCs

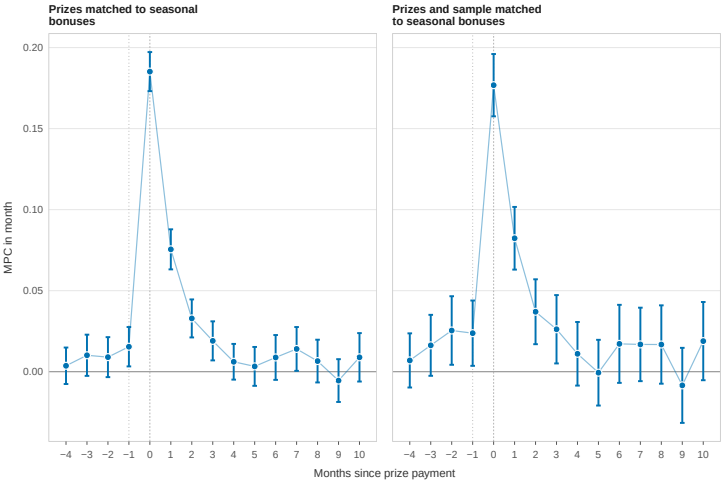
# Lottery wins: frequent, often bonus-sized, and randomly timed

- About 85% of Spanish adults play the state lottery; El Gordo is the world's largest prize pool.
- **30,556 wins** in our data (repeat winners and top 1% excluded); median prize €3,000.
- Identification: random timing among ever-winners (Borusyak et al. 2024 estimator).

| Group | N | Min (euros) | Mean (euros) | Median (euros) | Max (euros) |
|---|---|---|---|---|---|
| All lottery winners | 30,556 | 2 | 14,474 | 3,000 | 328,000 |
| Prize quintile 1 | 6,137 | 2 | 294 | 250 | 574 |
| Prize quintile 2 | 6,136 | 577 | 1,664 | 2,000 | 2,376 |
| Prize quintile 3 | 8,941 | 2,376 | 4,167 | 3,557 | 6,000 |
| Prize quintile 4 | 3,332 | 6,001 | 9,508 | 8,645 | 18,066 |
| Prize quintile 5 | 6,010 | 18,094 | 60,118 | 35,000 | 328,000 |
| Prizes matched to seasonal bonuses | 11,029 | 658 | 2,281 | 2,301 | 4,062 |
| Prizes and sample matched to seasonal bonuses | 3,844 | 667 | 2,272 | 2,269 | 4,052 |

‣ Estimator ‣ Winner characteristics

# MPCs from lottery wins size-matched to seasonal bonuses



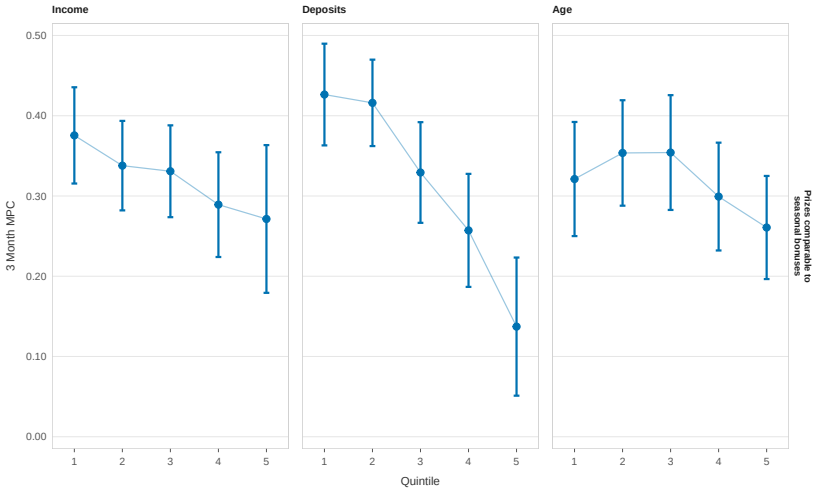

Prizes €657–4,065

1m MPC: **0.18**

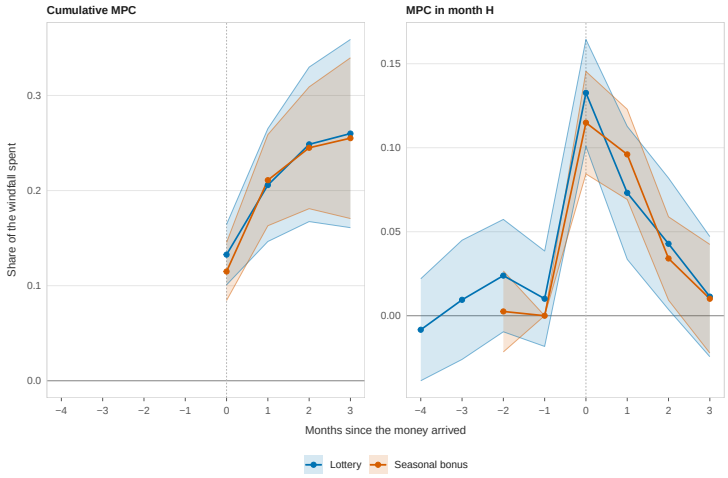
3m MPC: **0.30**

▸ Estimator ▸ MPC by prize size ▸ BJS vs FHN ▸ Never-winners ▸ Cumulative ▸ By category

# Lottery MPC heterogeneity



# Same households: bonus and lottery MPCs are indistinguishable




1,078 households both:

- Get seasonal bonus
- Later win similar prize

3m MPC = 0.25 **for both**

‣ Winner characteristics

## Our facts about MPCs

1. 3-month MPC out of seasonal bonuses: **0.21**.
2. 3-month MPC out of size-matched lottery wins: **0.30**.
3. For the same households: **indistinguishable**.
4. Both are **short-lived**: revert within 3 months.
5. Both decline steeply with **deposits**.
6. But both **above 0.1** in top quintile of deposit and income distributions.

‣ Cash holdings ‣ Wealth in Spain ‣ Browning & Collado

# Buffer Stock Models

# A standard buffer-stock model, plus a bonus every six months

$$\max_{\{c_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \quad \text{s.t.} \quad c_t + a_{t+1} = R\, a_t + \kappa_{j,g}\, z_t, \qquad \boldsymbol{a_{t+1} \geq 0}$$

$$\ln z_{t+1} = \rho \ln z_t + \varepsilon_{t+1}, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad \beta R < 1$$

- Standard one-asset model (Deaton 1991; Carroll 1997).
- One addition, two pay groups with the same present value of income: **flat**, $\kappa = 1$; **bonus**, $\kappa = 2w^*$ in June and December, $w^*$ otherwise. Fully anticipated.
- **MPC-IS**: the standard MPC, the response to an unanticipated windfall of bonus size:

$$\text{MPC-IS}(a, z, j) = \frac{c^*(a + zw^*, z, j) - c^*(a, z, j)}{w^*}$$

- **MPC-SB**: the same calculation as in our data, a difference in group means:

$$\text{MPC-SB} = \frac{(\bar{C}_T^b - \bar{C}_{T-1}^b) - (\bar{C}_T^f - \bar{C}_{T-1}^f)}{w^*}$$

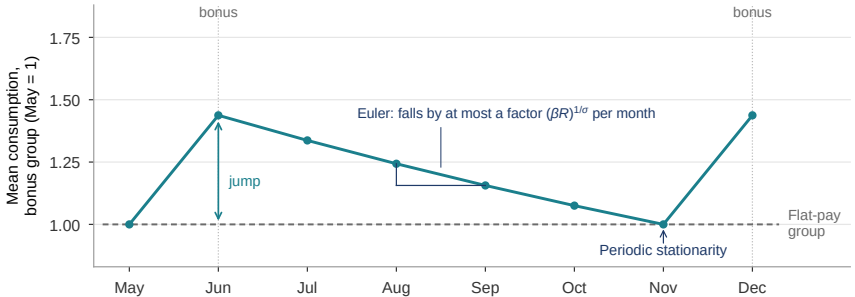
‣ Seasonal prefs, shocks

# In this model there is an analytical bound on the seasonal bonus MPC

$$u'(c_t) \;\geq\; \beta R\, E_t\left[u'(c_{t+1})\right] \qquad \text{Euler inequality}$$

$$\Longrightarrow \quad \bar{C}_{t+1} \;\geq\; (\beta R)^{1/\sigma}\, \bar{C}_t \qquad \text{rate of fall}$$

$$\Longrightarrow \quad \text{MPC-SB}(3m) \;\leq\; \frac{\bar{C}^b_{T-1}}{w^*} \sum_{h=0}^{2} \left((\beta R)^{-(5-h)/\sigma} - 1\right) \qquad \text{+ periodic stationarity}$$




$\sigma = 1.5$, $R = 1.02$, $\beta = 0.95$ $\Rightarrow$ 3-month MPC-SB $\leq$ **0.025**

3-month MPC-SB = 0.21 $\Rightarrow$ $\beta \leq$ **0.76**

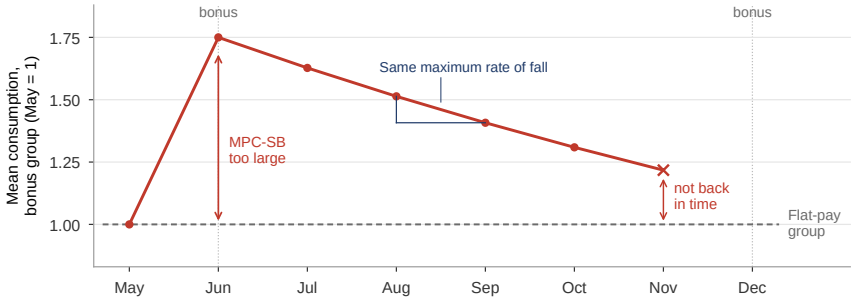

‣ Euler: proof ‣ Bound: proof ‣ Table ‣ Robustness ‣ Categories ‣ Why so small?

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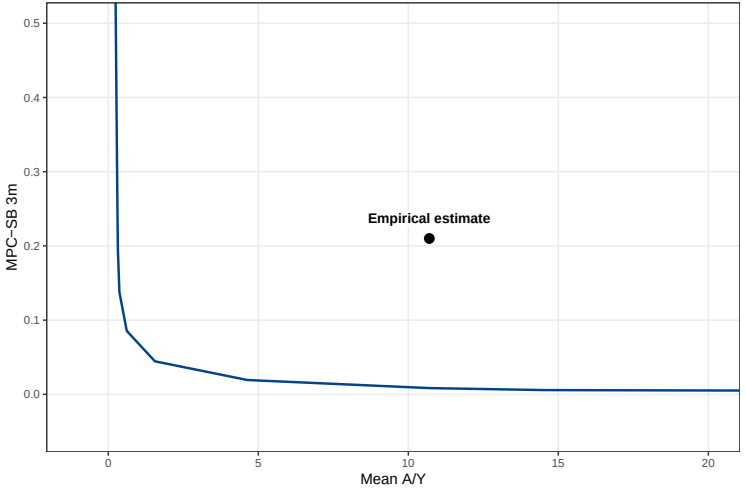



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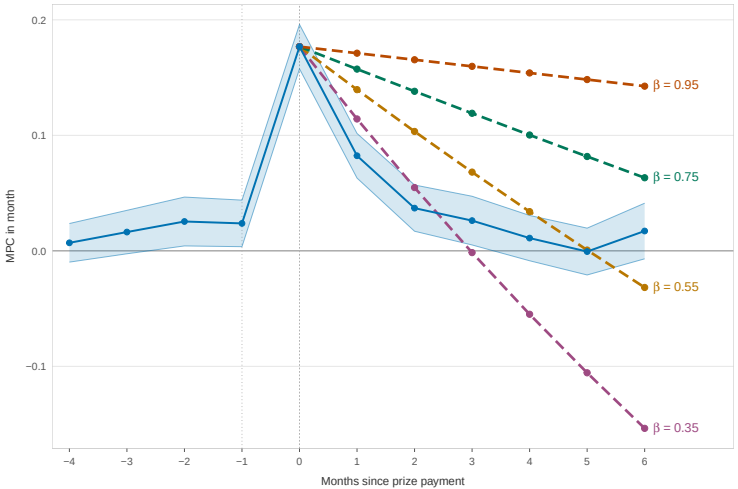

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# Simulating the model implies a severe trade-off between asset accumulation and MPC-SB




▸ All panels ▸ Seasonal prefs ▸ Results table ▸ Median wealth ▸ Wealth in Spain

# Corollary: a restriction on the shape of the iMPC path




▸ Proposition 1

## This finding is robust to a range of model variations

- **Seasonal preferences, expense shocks:** more saving ahead of high-need months, so fewer constrained households and a lower MPC-SB. ‣ Simulations
- **Fat-tailed income risk** (GKOS): more wealth; the MPC-SB falls further while the MPC-IS rises. ‣ Simulations
- **Higher risk aversion:** a higher $\sigma$ tightens the bound. ‣ Bound at $\sigma = 10$
- **Two-asset models:** explain high MPCs with high *total* wealth, not with moderate *liquid* wealth.
- **Heterogeneous discount factors:** need ~50% hand-to-mouth, or a fringe with an MPC near 1. In the data the MPC is above 0.1 everywhere.

‣ Results table

# Behavioural Models

# Mental accounting: two accounts, defined by when income arrived

- **Mental accounting:** people treat resources from different sources as **non-fungible** (Thaler 1985; Shefrin & Thaler 1988).
- **Macro version: past** and **present** income are non-fungible. Households dislike dipping into their savings (Graham & McDowall 2025; Gimeno-Ribes 2025).

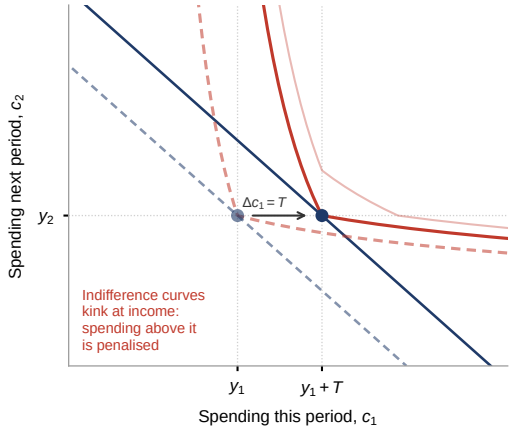
| Account | What is in it | A marginal euro is worth |
|---|---|---|
| Current income | labour income $y_t$, incl. bonuses and lottery wins | $u'(c)$ |
| Savings | previously accumulated income $a_t$ | $(1-\psi)\,u'(c)$ |

- **Spend current income first:** $c_t^y = \min\{c_t, y_t\}$, $c_t^a = \max\{c_t - y_t, 0\}$, so

$$v'(c_t) \;=\; \begin{cases} u'(c_t) & \text{if } c_t \le y_t \\ (1-\psi)\,u'(c_t) & \text{if } c_t > y_t \end{cases}$$

‣ Motivation ‣ Welfare cost ‣ Present bias

# How does mental accounting affect MPCs?




**MPCs increase**

Households set $c = y$ when

$$(1-\psi)^{1/\sigma} c^* \;\le\; y \;\le\; c^*$$

A bonus raises $y$ and shifts the kink by the same amount, so spending jumps.

**Mechanism remains at high wealth**

$y$ is persistent, so $c^*$ rises with $y$: some wealthy households are still at the kink.

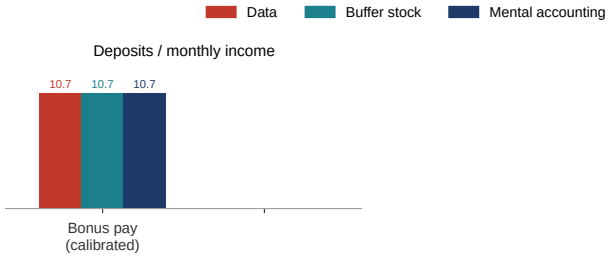
**Motive to accumulate wealth remains**

The penalty cancels over the 6-month cycle:

$$\beta(1-\psi) \cdot \beta^4 \cdot \frac{\beta}{1-\psi} \;=\; \beta^6$$

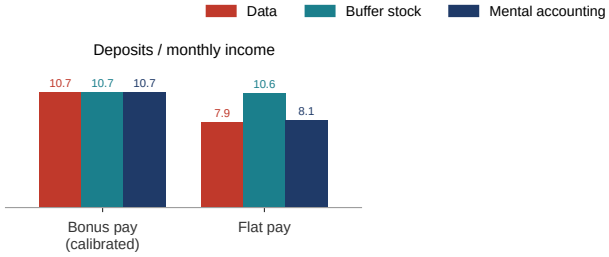
▸ Diagram ▸ Effective discounting

# Mental accounting matches both MPCs and wealth in one calibration



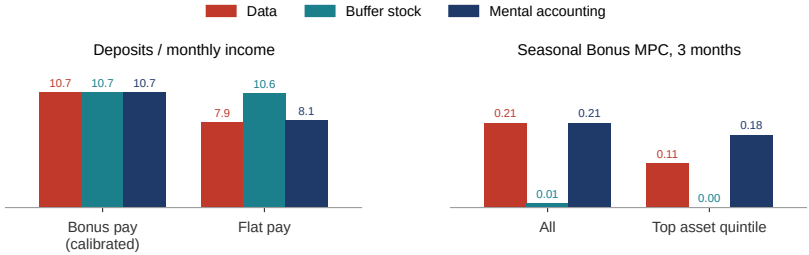

▸ Results table ▸ All panels ▸ Median wealth ▸ Model visualized

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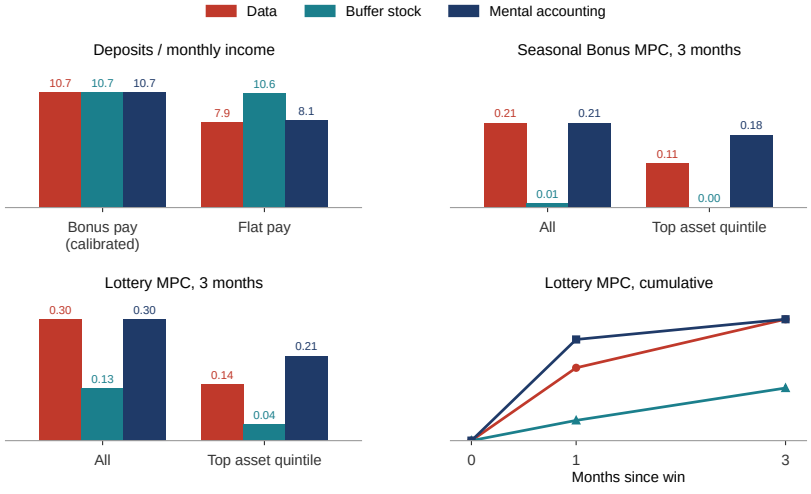

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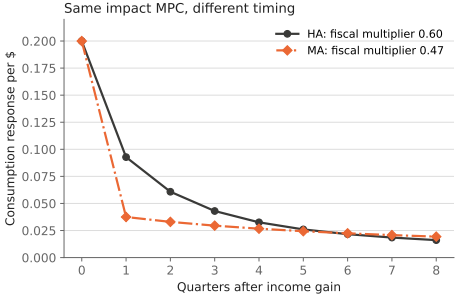
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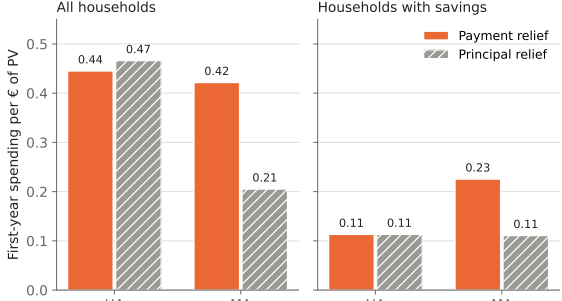



▸ Results table ▸ All panels ▸ Median wealth ▸ Model visualized

# Macroeconomic implications of mental accounting

- **Fiscal policy:** for a given impact MPC, fiscal multipliers are lower because MPCs are more front-loaded.
- **Multipliers remain substantial even in more equal wealth distributions**.
- **Policies targeting flows rather than stocks are more powerful.** In HA, principal relief is preferred; in MA, payment relief is much more stimulating.




‣ Model and calibration ‣ Fiscal policy ‣ Debt relief and credit

## Conclusion

- MPCs are **substantial** out of fully anticipated, recurring bonuses, and similar magnitude to MPCs out of lotteries.
- Buffer-stock models cannot reproduce this: the Euler inequality makes responses **persistent**, and bonus MPCs **negligible**.
- **Mental accounting** models can match the level and shape of both MPCs.
- Agenda going forward: more work using this granular microdata — retirement, monetary transmission, and saving.

# End